THE NORMALIZATION OF THE BASE IN A DISJOINT UNION IS THE DISJOINT UNION OF THE NORMALIZATIONS

THE STACKS PROJECT AUTHORS

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Editorial note. Statement span 14095-14101 and proof span 14103-14113 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof ends with the author's parenthesis '(some details omitted)' for the identification of the two factors of the integral closure; that phrase is kept verbatim and what it leaves to the reader is the routine verification, while the mechanism of the proof (integrality of the idempotents) is stated in the counted span.

Allowed standard prerequisites (editorial).

- The normalization of a scheme in a quasi-compact quasi-separated morphism, defined as the relative spectrum of the integral closure of the structure sheaf in the pushforward of the structure sheaf of the source. Used at statement lines 14312-14318 and implicitly throughout the proof. Names the two objects being compared and reduces the claim to a statement about integral closures.
- The integral closure of a ring in a product of rings is the product of the integral closures in the factors, via the idempotents of the product. Used at proof lines 14320-14328. Supplies the algebraic fact that the proof invokes and sketches.
- Idempotents of a ring are integral over any subring. Used at proof lines 14324-14326. Is the key observation that forces the integral closure inside a product to be a product.

Lemma 0.1. *Let $f : Y \to X$ be a quasi-compact and quasi-separated morphism of schemes. Suppose that $Y = Y_1 \amalg Y_2$ is a disjoint union of two schemes. Write $f_i = f|_{Y_i}$. Let X_i' be the normalization of X in Y_i. Then $X_1' \amalg X_2'$ is the normalization of X in Y.*

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Proof. In terms of integral closures this corresponds to the following fact: Let $A \to B$ be a ring map. Suppose that $B = B_1 \times B_2$. Let A_i' be the integral closure of A in B_i. Then $A_1' \times A_2'$ is the integral closure of A in B. The reason this works is that the elements $(1, 0)$ and $(0, 1)$ of B are idempotents and hence integral over A. Thus the integral closure A' of A in B is a product and it is not hard to see that the factors are the integral closures A_i' as described above (some details omitted). $\square$